

From Groups to Modules

A pedagogical tour of five algebraic structures

Roadmap

We will build five structures, in this order:

$$\text{group} \longrightarrow \text{ring} \longrightarrow \text{field} \longrightarrow \text{vector space} \longrightarrow \text{module}.$$

Each arrow adds either an operation, an axiom, or a notion of “scalar multiplication.” The last arrow is the only *generalization* in the chain: a module is what you get when you take a vector space and allow the scalars to come from a ring instead of a field. That step trips most people up the first time, so we will spend extra care on it.

1 Group

The idea. A group captures the bare minimum needed for an operation that can be *undone*. Think: integers under addition (you can subtract), or the symmetries of a square (you can rotate back).

Definition. A *group* $(G, \cdot)$ is a set G with a binary operation $\cdot : G \times G \rightarrow G$ satisfying:

- (G1) *Associativity:* $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$.
- (G2) *Identity:* there is an element $e \in G$ with $e \cdot a = a \cdot e = a$ for all a .
- (G3) *Inverses:* for each $a \in G$ there is some $a^{-1} \in G$ with $a \cdot a^{-1} = a^{-1} \cdot a = e$.

If, in addition, $a \cdot b = b \cdot a$ always holds, the group is called *abelian*.

Example. $(\mathbb{Z}, +)$ is an abelian group: identity 0, inverse of n is $-n$.

Example. The six symmetries of an equilateral triangle (three rotations and three reflections) form a non-abelian group, traditionally written S_3 .

2 Ring

The idea. A group has *one* operation. But integers, polynomials, and matrices have *two* — addition and multiplication — and the two interact through distributivity. A ring axiomatizes this two-operation world.

Definition. A *ring* $(R, +, \cdot)$ is a set with two binary operations such that:

- (R1) $(R, +)$ is an abelian group (its identity is written 0).
- (R2) $\cdot$ is associative.
- (R3) *Distributivity:* $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

We will assume rings have a multiplicative identity 1 (such rings are called *unital*).

Example. $\mathbb{Z}$ with the usual $+$ and $\times$ is the prototypical ring.

Example. Polynomials $\mathbb{R}[x]$ form a ring. So do $n \times n$ matrices over $\mathbb{R}$ (non-commutative when $n \geq 2$).

What's missing? In a ring you can add, subtract, and multiply — but you cannot generally *divide*. There is no element $\frac{1}{2}$ in $\mathbb{Z}$. The next step fixes this.

3 Field

The idea. A field is a ring in which division by anything nonzero is allowed. $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are the canonical examples.

Definition. A *field* $(F, +, \cdot)$ is a commutative unital ring in which every nonzero $a \in F$ has a multiplicative inverse $a^{-1} \in F$ (so $a \cdot a^{-1} = 1$).

Equivalently: $(F, +)$ and $(F \setminus \{0\}, \cdot)$ are both abelian groups, glued by distributivity.

Example. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are fields.

Example. For each prime p , the integers modulo p , written $\mathbb{F}_p$, form a finite field with p elements.

Remark. Why *prime*? In $\mathbb{Z}/6\mathbb{Z}$ the element 2 has no inverse: there is no x with $2x \equiv 1 \pmod{6}$. Primality is what guarantees every nonzero element is invertible. So $\mathbb{Z}/6\mathbb{Z}$ is a ring but not a field — keep this example in mind, it will return.

4 Vector space

The idea. The setting of linear algebra: things you can add together and *scale by numbers*. The numbers (“scalars”) come from a field.

Definition. Let F be a field. A *vector space over F* is an abelian group $(V, +)$ together with an operation

$$F \times V \rightarrow V, \quad (\alpha, v) \mapsto \alpha v,$$

called *scalar multiplication*, satisfying for all $\alpha, \beta \in F$ and $u, v \in V$:

$$(V1) \quad \alpha(u + v) = \alpha u + \alpha v$$

$$(V2) \quad (\alpha + \beta)v = \alpha v + \beta v$$

$$(V3) \quad (\alpha\beta)v = \alpha(\beta v)$$

$$(V4) \quad 1 \cdot v = v$$

Example. $\mathbb{R}^n$ with componentwise addition and scaling is the standard vector space over $\mathbb{R}$.

Example. Continuous functions $C([0, 1], \mathbb{R})$ form an (infinite-dimensional) vector space over $\mathbb{R}$.

The payoff. Because the scalars form a field, every vector space has a *basis*, and any two bases have the same size (the *dimension*). This is what makes linear algebra so clean.

5 Module — the generalization

The natural question. Look back at the vector space definition. Where did we use the fact that F was a *field*, and not merely a ring? Answer: nowhere in the four axioms. We only used “field-ness” afterwards, when we wanted bases to exist.

So: what if we keep the four axioms but let the scalars come from a mere ring R ? That object is called a *module*.

Definition. Let R be a unital ring. A *module over R* (or an *R -module*) is an abelian group $(M, +)$ together with an operation

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm,$$

satisfying for all $r, s \in R$ and $m, n \in M$:

$$(M1) \quad r(m + n) = rm + rn$$

$$(M2) \quad (r + s)m = rm + sm$$

$$(M3) \quad (rs)m = r(sm)$$

$$(M4) \quad 1 \cdot m = m$$

This is literally the vector space definition with the letter F replaced by the letter R . Nothing else changes.

Reading the phrase “module over R ”

The phrase “ M is a module over the ring R ” just names which ring is supplying the scalars. The letter R is a placeholder — substitute any specific ring you like.

<i>vector space over $\mathbb{R}$</i>	scalars are real numbers
<i>$\mathbb{Z}$-module</i>	scalars are integers
<i>module over R</i>	scalars come from some unspecified ring R

So a “ $\mathbb{Z}$ -module” is not a different beast from “a module over a ring R .” It is the special case where $R = \mathbb{Z}$.

Worked example: every abelian group is a $\mathbb{Z}$ -module

This is the example that makes “scalars from a ring” click.

Take *any* abelian group $(A, +)$. We will multiply elements of A by integers in the only way that makes sense, and check that this turns A into a $\mathbb{Z}$ -module — automatically, for free, with no extra structure required.

For $n \in \mathbb{Z}$ and $a \in A$, define

$$n \cdot a = \begin{cases} \underbrace{a + a + \cdots + a}_{n \text{ copies}} & n > 0, \\ 0 & n = 0, \\ \underbrace{(-a) + (-a) + \cdots + (-a)}_{|n| \text{ copies}} & n < 0. \end{cases}$$

This is just *repeated addition*. The four module axioms (M1)–(M4) are now forced — they follow from associativity and commutativity of $+$ in A . So *every abelian group becomes a $\mathbb{Z}$ -module, in exactly one natural way*.

Example (Concrete walkthrough). Let $A = \mathbb{Z}/6\mathbb{Z} = \{[0], [1], [2], [3], [4], [5]\}$, an abelian group under addition modulo 6. Treat it as a $\mathbb{Z}$ -module. Then:

$$\begin{aligned} 3 \cdot [2] &= [2] + [2] + [2] = [6] = [0], \\ (-1) \cdot [2] &= [-2] = [4], \\ 7 \cdot [3] &= [21] = [3]. \end{aligned}$$

Here 3, -1 , 7 are the *scalars* (elements of $\mathbb{Z}$); $[2], [3]$ are the *module elements* (elements of $\mathbb{Z}/6\mathbb{Z}$). The integer 3 acts on $[2]$ by “add $[2]$ to itself three times.”

Remark (The moral). “Abelian group” and “ $\mathbb{Z}$ -module” are the same concept in two languages. The module language just makes the implicit “repeated addition” structure visible as an explicit scalar action.

What is lost without a field?

In $\mathbb{Z}/6\mathbb{Z}$ as a $\mathbb{Z}$ -module, we saw $3 \cdot [2] = [0]$. There is no way to “divide by 3” and recover $[2]$, because $3 \in \mathbb{Z}$ has no multiplicative inverse in $\mathbb{Z}$.

In a vector space (scalars in a field) this can never happen: if $\alpha v = 0$ with $\alpha \neq 0$, multiply by α^{-1} to conclude $v = 0$.

The consequence: modules need not have bases. The $\mathbb{Z}$ -module $\mathbb{Z}/6\mathbb{Z}$ has no basis at all — every element is killed by 6. When a module *does* have a basis, it is called *free*.

Summary

Structure	What it has	Key new feature
Group	one operation	inverses (can undo)
Ring	two operations $(+, \cdot)$	distributivity
Field	two operations $(+, \cdot)$	every nonzero scalar invertible
Vector space	$+$ and scaling by a field	bases exist; dimension defined
Module	$+$ and scaling by a ring	generalizes vector space; basis may not exist

The hierarchy is one of progressively richer *scalars*:

$$\text{abelian group} = \mathbb{Z}\text{-module} \subset R\text{-module} \supset F\text{-vector space}.$$

The last $\supset$ is the punchline: vector spaces are the special, well-behaved case of modules — the case where the scalars happen to form a field.